

BRANDON M. DURAN

PhD Candidate, Scripps Institution of Oceanography

bmduran@ucsd.edu

ORCID: 0000-0001-5427-8700

RESEARCH INTERESTS

aerosol-cloud interactions, aerosol radiative forcing, circulation responses to aerosol forcing, satellite simulators

EDUCATION

Scripps Institution of Oceanography, University of California San Diego 2022 - Present
Masters in Earth Science (2024), Ph.D. expected in 2027

Georgetown University 2018-2022
BA in Physics, Computer Sciences

PEER REVIEWED PUBLICATIONS

4. **Duran, B.M.**, Lutsko, N.J., and Wall, C.J. Aerosol-Ice-Cloud Interactions in a Perturbed Parameter Ensemble (2026) *Journal of Climate*.
3. Michibata, T., Wall, C.J., Hirota, N., **Duran, B.M.**, and Nozawa, T. Recent Advances in the Observation and Modeling of Aerosol-Cloud Interactions, Cloud Feedbacks, and Earth's Energy Imbalance: A Review (2025). *Current Pollution Reports*.
<https://doi.org/10.1007/s40726-025-00382-6>.
2. Wall, C.J., Paynter, D., Qin, Y., Debolskiy, M., Duffy, M.L., Michibata, T., **Duran, B.M.**, Lutsko, N.J., Ma, P.L., Medeiros, B., Storelvmo, T., and Zhao, M. Decomposing Cloud Radiative Feedbacks by Cloud-Top Phase (2025). *Journal of Climate*. <https://doi.org/10.1175/JCLI-D-24-0538.1>.
1. **Duran, B.M.**, Wall, C.J., Lutsko, N.J., Michibata, T., Ma, P.L., Qin, Y., Duffy, M.L., Medeiros, B., and Debolskiy, M. A new method for diagnosing effective radiative forcing from aerosol-cloud interactions in climate models (2025). *Atmospheric Chemistry and Physics*.
<https://doi.org/10.5194/acp-25-2123-2025>.

SUBMITTED MANUSCRIPTS

1. Wynn, S., Watson-Parris, D., **Duran, B.M.**, and Mülmenstädt, C.J. Base-state dependence of parameter sensitivity for the Twomey effect in a perturbed parameter ensemble (preprint, 2026) *Atmospheric Chemistry and Physics*.

AWARDS RECEIVED

2022	National Science Foundation Graduate Research Fellow
2020-2022	National Hispanic Scholar
2021	AGU Student Travel Grant Recipient

CONFERENCE PRESENTATIONS

2026 Oral: 2026 CESM Workshop
2026 Oral: Weather & Climate Livestream
2026 Oral: CESM Winter Working Group Meeting
2025 Poster: Gordon Research Conference, Radiation & Climate at Bates College
2025 Oral: 2025 CESM Workshop
2025 Invited Talk: Equilibrium Climate Sensitivity and Cloud Feedback Virtual Symposium
2024 Poster: AGU, Washington, D.C.
2024 Poster: Micro2Macro Workshop, University of Wyoming
2024 Oral: CFMIP Conference at Boston University

ACADEMIC TEACHING EXPERIENCE

2026 Teaching assistant for undergraduate course *Dynamics of Atmos & Climate*
 Taught lectures and oversaw grading.
2025 Teaching assistant for undergraduate course *The Atmosphere*. Oversaw grading and
 office hours for 90+ students. (UC San Diego)
2021 Teaching assistant for undergraduate physics courses *Mechanics* and
Electromagnetic Phenomena, Georgetown University

SERVICE

2026- Student Transportation Advisory Committee, member
2024- Climate Sciences representative and senior co-chair
 of the Scripps Graduate Student Council
2024- Co-organizer of SIO Climate Journal Club
2023-24 Co-organizer of the Climate, Atmospheric Sciences and Physical
 Oceanography (CASPO) seminar series